Progress Report — First and Second-Stage Results

February 2026

Research Question

Is the sector-level allocation of BNDES lending across municipalities GDP-optimal?

- BNDES disburses $\sim$3% of GDP annually in subsidized industrial lending
- Existing literature (Lane 2025, Juhász 2018) shows IP "works" for targeted sectors
- **But**: does the policymaker target the *right* sectors?
- We derive and test the **first-order condition for optimality**: at an interior optimum, marginal reallocations across sectors have zero effect on GDP

Contribution: Move from "does IP work?" to "is IP *optimal*?" — and if not, identify which sectors are over/under-supported

Theory: Local Optimality Test

Sector shares $s_m \in \Delta^{J-1}$, municipal output $Y_m = Y_m(s_m)$.

Directional derivative condition — at any interior optimum:

$$\left.\frac{dY_m}{d\varepsilon}\right|_{\varepsilon=0} = \nabla_{s_m} Y_m \cdot v_m = 0 \quad \text{for all } v_m \perp \mathbf{1}$$

Empirical projection (within-simplex, $\mathbf{1}'\Delta s_{mt} = 0$):

$$\Delta Y_{m,t+h} = \beta' \Delta s_{mt} + u_{m,t+h}$$

Null hypothesis (optimality):

$$H_0 : \quad \beta = \mathbf{0}$$

- Model-free: valid under any smooth objective
- $\hat{\beta}_j > 0$: sector j under-supported $\implies$ tilt more toward j
- $\hat{\beta}_j < 0$: sector j over-supported

Identification: Shift-Share IV from Political Alignment I

Δs_{mt} is endogenous $\implies$ need instruments that shift the sector mix exogenously.

Shift — Political alignment turnover:

- When party p wins office $\ell \in \{\text{mayor, governor, president}\}$, BNDES credit tilts toward p-affiliated firms (Lazzarini et al. 2015, Carvalho 2014)
- $\Delta\mathrm{Align}^{\ell}_{mtp}$: party-$p$ alignment shock in municipality m, year t

Share — Baseline sectoral exposure to each party:

$$w_{mjp} \equiv \frac{\sum_{f \in (m,j)} L_{fp,t_0}}{\sum_{f \in (m,j)} N_{f,t_0}}$$

where L_{fp} = number of owners of firm f affiliated to party p; N_f = total number of owners (affiliated and unaffiliated) of firm f.

Identification: Shift-Share IV from Political Alignment II

Instrument (sector-specific, within municipality):

$$Z^{\ell}_{mjt} \equiv \sum_{p} w_{mjp} \cdot \Delta \mathrm{Align}^{\ell}_{mtp}$$

Coverage: 82.3% of firm-muni-year rows have owner affiliation data ($N_f / F_f \approx 1.03$, near one owner per firm). At the sector-muni-year level, 74.5% of cells have ≥ 1 affiliated owner; the remaining 25.5% receive $Z_{mjt} = 0$. These zero-instrument cells account for only 1.8% of BNDES disbursement value.

Estimation Strategy

First stage (muni × sector × year):

$$\Delta s_{mjt} = \Pi Z_{mjt} + \alpha_{mj} + \alpha_{mt} + \eta_{mjt}$$

- α_{mj}: muni × sector FE α_{mt}: muni × year FE
- Muni×year FE absorbs the **aggregate** political alignment shock
- Identification from **cross-sector variation** in party exposure w_{mjp}

Second stage (muni × year, reduced form with sector-specific instruments):

$$\ln Y_{mt} = \sum_{j \neq j_0} \beta_j \widehat{\Delta s}_{mjt} + \color{red}{\psi \text{BNDES}_{mt}} + \delta_m + \delta_t + u_{mt}$$

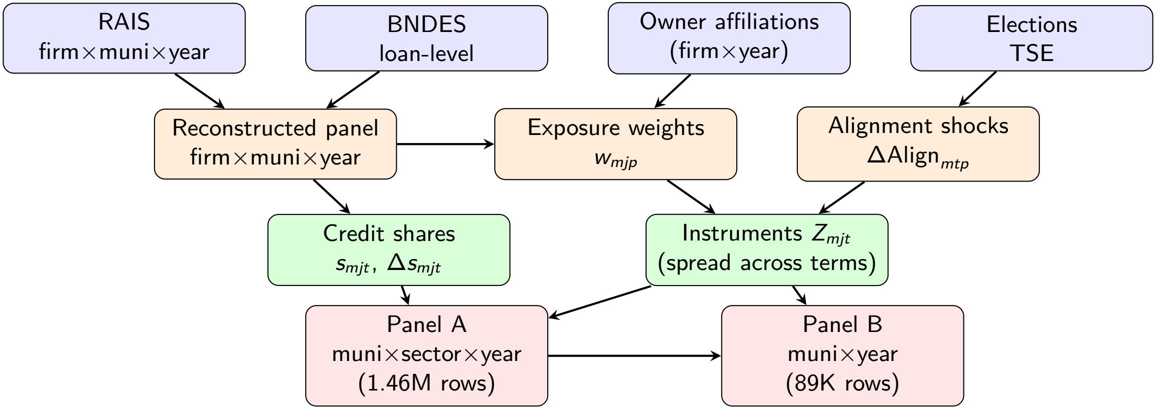
- β_j: GDP effect of reallocating 1pp from j_0 to sector j
- Control for BNDES_{mt}?: if instrument changes both composition and scale, controlling for scale may introduce bias.
- **Optimality test**: joint Wald $H_0 : \beta = \mathbf{0}$

Data Sources

Source	Content	Coverage
BNDES transparency portal	Loan-level disbursements	2002–2024
RAIS (MTE)	Employer-employee, CNAE sector	2002–2017
TSE / Colonnelli et al.	Worker affiliation & donations (firm-level)	2002–2019
IBGE PIB Municipal	Municipality GDP	2002–2017
IBGE Population	Municipality population	1991–2025
IPCA (IBGE)	Price index for deflation	Base 2018
SICONFI / basedosdados	Municipal transfers (placebo)	2002–2017

Geographic unit: 5,570 valid municipalities
Sector definition: 21 CNAE 2.0 sections (letters A–U; see distribution)

Pipeline Architecture



Key Construction Decision: Instrument Timing

Problem: Alignment shocks $\Delta\text{Align}^{\ell}_{mtp}$ are only non-zero at inauguration years.

- Mayors: 2005, 2009, 2013, 2017
- Governors/Presidents: 2003, 2007, 2011, 2015
- $\implies$ instruments zero for ∼70% of panel

Solution: Spread each shock across the full 4-year electoral term.

- Political alignment persists until the next election
- Mayor inauguration 2009 → shock applied to 2009, 2010, 2011, 2012
- Gov/Pres inauguration 2011 → shock applied to 2011, 2012, 2013, 2014

After spreading: Non-zero instruments for ∼65% of muni-years (coalition instruments).
Coverage: all years 2003–2017.

First Stage (Coalition): Single-Tier Instruments

	(1) Mayor	(2) Governor	(3) President
$Z^{\text{mayor}}_{\text{coalition}}$	0.0282^{***}		
	(0.0080)		
$Z^{\text{gov}}_{\text{coalition}}$		-0.0039	
		(0.0095)	
$Z^{\text{pres}}_{\text{coalition}}$			-0.0225
			(0.0151)
Fixed Effects	$\alpha_{mj} + \alpha_{mt}$	$\alpha_{mj} + \alpha_{mt}$	$\alpha_{mj} + \alpha_{mt}$
Clustering	muni + CNAE	muni + CNAE	muni + CNAE
Observations	1,372,575	1,372,575	1,372,575
F-statistic	**12.4**	0.17	2.23

- **Mayor alignment drives sector reallocation**: $F = 12.4$, passes Stock-Yogo threshold
- Interpretation: 1 s.d. mayor alignment shock $\rightarrow$ 2.8pp shift in sector share

First Stage (Party): Single-Tier Instruments

	(1) Mayor	(2) Governor	(3) President
$Z^{\text{mayor}}_{\text{party}}$	0.0476^{***}		
	(0.0166)		
$Z^{\text{gov}}_{\text{party}}$		-0.0314^{***}	
		(0.0101)	
$Z^{\text{pres}}_{\text{party}}$			0.0601^{*}
			(0.0337)
Fixed Effects	$\alpha_{mj} + \alpha_{mt}$	$\alpha_{mj} + \alpha_{mt}$	$\alpha_{mj} + \alpha_{mt}$
Clustering	muni + CNAE	muni + CNAE	muni + CNAE
Observations	1,372,575	1,372,575	1,372,575
F-statistic	**8.24**	**9.65**	3.19

- **Both mayor and governor individually significant** (unlike coalition)
- Party alignment is a more precise measure: fewer affiliated firms per party → sharper cross-sector variation

2002-fixed baseline

First Stage (Coalition): Combined Instruments

	(1) M+G	(2) M+P	(3) M+G+P
$Z^{\text{mayor}}_{\text{coalition}}$	0.0287^{***}	0.0251^{***}	0.0252^{***}
	(0.0069)	(0.0065)	(0.0054)
$Z^{\text{gov}}_{\text{coalition}}$	0.0024		0.0006
	(0.0084)		(0.0097)
$Z^{\text{pres}}_{\text{coalition}}$		-0.0184	-0.0184
		(0.0141)	(0.0147)
Fixed Effects	$\alpha_{mj} + \alpha_{mt}$	$\alpha_{mj} + \alpha_{mt}$	$\alpha_{mj} + \alpha_{mt}$
Clustering	muni + CNAE	muni + CNAE	muni + CNAE
Observations	1,372,575	1,372,575	1,372,575
Joint F	**14.1**	8.2	**9.4**
p-value	< 0.001	< 0.001	< 0.001

- Mayor instrument stable across specifications (0.025–0.029)

First Stage (Party): Combined Instruments

	(1) M+G	(2) M+P	(3) M+G+P
$Z^{\text{mayor}}_{\text{party}}$	0.0454^{**}	0.0480^{**}	0.0459^{**}
	(0.0164)	(0.0169)	(0.0168)
$Z^{\text{gov}}_{\text{party}}$	-0.0266^{**}		-0.0239^{**}
	(0.0098)		(0.0097)
$Z^{\text{pres}}_{\text{party}}$		0.0611^{*}	0.0578
		(0.0348)	(0.0359)
Fixed Effects	$\alpha_{mj} + \alpha_{mt}$	$\alpha_{mj} + \alpha_{mt}$	$\alpha_{mj} + \alpha_{mt}$
Clustering	muni + CNAE	muni + CNAE	muni + CNAE
Observations	1,372,575	1,372,575	1,372,575
Joint F	**6.40**	5.71	**5.34**
p-value	0.002	0.003	0.001

- Governor negative sign: winning party's sector loses share (crowd-out from other sectors?)

Coalition vs. Party: Comparison I

	Coalition			Party		
	Coef	SE	F	Coef	SE	F
Single-tier (muni$\times$year FE):						
Mayor	0.028^{***}	(0.008)	12.4	0.048^{***}	(0.017)	8.2
Governor	-0.004	(0.010)	0.2	-0.031^{***}	(0.010)	9.7
President	-0.023	(0.015)	2.2	0.060^{*}	(0.034)	3.2
Combined M+G+P:						
Joint F		**9.4**			5.3	
p-value		< 0.001			0.001	
Year FE robustness (M+G+P):						
Joint F		**60.4**			28.5	

- **Coalition**: stronger joint F (9.4), concentrated in mayor tier

Coalition vs. Party: Comparison II

- **Party**: unlocks governor channel ($F = 9.7$), larger individual coefficients
- President sign flips: coalition negative vs. party positive
- Take-away: coalition preferred for joint F; party provides distinct variation at subnational tiers

FE robustness

First-Stage Interpretation and Second-Stage Assumptions

What the first stage shows:

- Mayor alignment turnover shifts BNDES sector shares within municipalities
- Sectors with more party-affiliated firm owners gain relatively more credit
- Effect is $\sim$2.8pp per unit alignment shock, robust across specifications

Key assumptions for second stage:

1. **Exclusion restriction**: Sector-specific Z_{mjt} affects GDP *only* through BNDES credit composition
   - Concern: transfers, procurement, regulation also respond to alignment shocks
   - Using sector-specific instruments helps: cross-sector variation in party exposure is less likely to correlate with transfers/procurement
   - Planned test: show sector Z does not predict municipal transfers
2. **Instrument relevance**: $F > 10$ (satisfied for mayor, borderline for all tiers)
3. **SUTVA / no spillovers**: reallocation in muni m doesn't affect muni m'
4. **Local approximation**: perturbations are "small" (median $|\Delta s| \approx 0.02$)

Second Stage: Reduced Form (Sector-Specific Instruments)

Specification	N	R^2	IVs	Wald F	p-value
Mayor	89,008	0.9455	18	1.47	0.0894
M+G	89,008	0.9456	36	2.18	$< 10^{-4}$
M+G+bndes_pc	89,008	0.9456	36	2.18	$< 10^{-4}$

- **Optimality null rejected**: M+G sector instruments jointly significant ($F = 2.18$, $p < 0.0001$)
- Mayor-only: marginal ($p = 0.09$); adding governor sector instruments sharpens rejection
- Adding bndes_{pc} (scale control) makes no difference: ≈ 0

Reduced Form: Notable Sector Coefficients

From the M+G specification (36 sector-specific instruments, $F = 2.18$):

Sector	Tier	Coef (SE)	Sign.
C — Manufacturing	Mayor	-0.080 (0.038)	**
P — Education	Mayor	$+0.390$ (0.096)	***
M — Professional	Mayor	$+0.255$ (0.147)	*
S — Other services	Mayor	-0.478 (0.280)	*
E — Water/waste	Governor	-0.462 (0.135)	***
I — Accommodation	Governor	$+0.372$ (0.145)	**
P — Education	Governor	$+0.671$ (0.197)	***
R — Arts/recreation	Governor	$+0.213$ (0.079)	***

- Manufacturing (C) **over-supported**: reallocating toward C reduces GDP
- Education (P) **under-supported**: both mayor and governor channels show positive effect
- Coefficients relative to dropped sector G (wholesale/retail)

Second Stage: Robustness

All robustness checks use M+G sector-specific instruments (36 IVs).

Specification	N	R^2	IVs	Wald F	p-value
2002-fixed baseline	89,008	0.9456	36	2.54	$< 10^{-4}$
Trimmed 1–99%	87,223	0.9466	36	2.04	0.0002
Cluster: muni+year	89,008	0.9456	36	1.71	0.0049

- **Optimality rejection robust** across all specifications
- 2002-fixed baseline: strongest rejection ($F = 2.54$, $p < 0.0001$)
- Survives two-way clustering ($p = 0.005$) and trimming ($p = 0.0002$)
- Transfer placebo not yet available (SICONFI data insufficient)

Second Stage: Interpretation

Key insight: Using sector-specific instruments isolates the BNDES composition channel.

- **First stage**: Mayor alignment $\rightarrow$ BNDES sector reallocation ($\hat{\Pi} = 0.028^{***}$, $F = 12.4$)
- **Reduced form**: Sector-specific instruments jointly predict GDP ($F = 2.18$, $p < 0.0001$)
- $\Longrightarrow$ **Optimality null rejected**: the current sectoral allocation is not GDP-optimal

Policy-relevant sectors:

- Manufacturing (C): $\pi < 0$ — over-funded relative to wholesale/retail
- Education services (P): $\pi > 0$ — under-funded, strong signal from both tiers
- Water/waste (E, gov channel): $\pi < 0$ — over-funded

Next: Vector 2SLS to estimate structural β_j (sector-specific GDP elasticities) and test which sectors drive the rejection.

Additional Design Choices

Choice	Description
Dropped sector	$j_0 = \text{G}$ (Wholesale/retail trade), largest mean share (18.2%). β_j is relative to j_0. Joint test invariant to choice.
Sparse sectors	T (Households as employers) and U (Extraterritorial) dropped: $<$100 nonzero instrument obs in 89K rows. Caused extreme leverage ($F = 82$ in 2002-fixed spec before dropping).
Baseline weights	Primary: cycle-specific (last year before election). Robustness: 2002-fixed (all cycles use first year of data).
Instrument def.	Currently: affiliated owner counts / total owners. Draft: employment-weighted within-sector party shares (TODO).

Next Steps

Immediate:

1. Aggregate some sectors and re-run regressions
2. Construct the employment-weighted instrument and re-run first and second stages
3. Run scalar 2SLS (Table 5: ΔHHI instrumented by aggregate Z)
4. Run vector 2SLS (Table 6: Δs_j instrumented by sector Z_j) for structural β_j
5. Transfer placebo (TODO: need to find data with better coverage)

Appendix: Sector Share Distribution (with zeros)

CNAE Section	Mean s_{mjt}	% Positive	N obs
G — Wholesale/retail trade	0.182	38.9%	89,152
C — Manufacturing	0.160	34.1%	85,008
H — Transport/storage	0.123	30.8%	88,656
A — Agriculture	0.036	11.1%	84,944
F — Construction	0.024	12.9%	85,424
J — Information/communication	0.024	8.3%	69,680
B — Mining	0.023	9.9%	52,480
N — Admin services	0.012	8.7%	84,944
D — Electricity/gas	0.009	2.3%	35,696
(12 sectors $<$ 1%)	*$<$0.007*	*$<$6%*	—

Total: 1,464,080 rows. 21 CNAE sections. 89.6% of cells have zero BNDES credit.
Dropped sector $j_0 = \text{G}$ (largest mean share).

Back to Data Sources

Appendix: First Stage — 2002-Fixed Baseline Instruments

	Coalition			Party		
	Mayor	Governor	President	Mayor	Governor	President
Coefficient	0.0138^{**}	-0.0093^{**}	0.0051	0.0102	-0.0208^{***}	0.0352
	(0.0056)	(0.0036)	(0.0066)	(0.0087)	(0.0056)	(0.0300)
F-stat	5.98	6.49	0.59	1.40	**13.9**	1.38
Coalition combined: $F = 6.73$, $p < 0.001$			*Party combined:* $F = 5.66$, $p < 0.001$			

- 2002-fixed baselines: clearly predetermined (first year of data)
- Coalition: mayor and governor individually significant
- Party: **governor is the strong instrument** ($F = 13.9$), mayor loses significance
- Pattern reversal from cycle-specific: party weights from 2002 are stale for mayor (shorter terms), but governor/president party structure more persistent

Back to First Stage

Appendix: First Stage — FE Robustness I

	Coalition		**Party**	
	Muni×yr FE	Year FE	Muni×yr FE	Year FE
Z^{mayor}	0.0252***	0.0228***	0.0459**	0.0430***
	(0.0054)	(0.0019)	(0.0168)	(0.0148)
Z^{gov}	0.0006	0.0003	−0.0239**	−0.0235**
	(0.0097)	(0.0083)	(0.0097)	(0.0091)
Z^{pres}	−0.0184	−0.0160	0.0578	0.0519
	(0.0147)	(0.0120)	(0.0359)	(0.0353)
Joint F (M+G+P)	**9.4**	**60.4**	5.3	**28.5**
p-value	< 0.001	< 0.001	0.001	< 0.001

- Coefficients robust across FE specifications within each alignment definition
- Year FE boosts F substantially for both definitions (as expected)

Appendix: First Stage — FE Robustness II

- **Conservative choice**: muni$\times$year FE as primary; coalition for strongest joint F
- Party instruments useful as robustness check and for overidentification tests

Back to Comparison